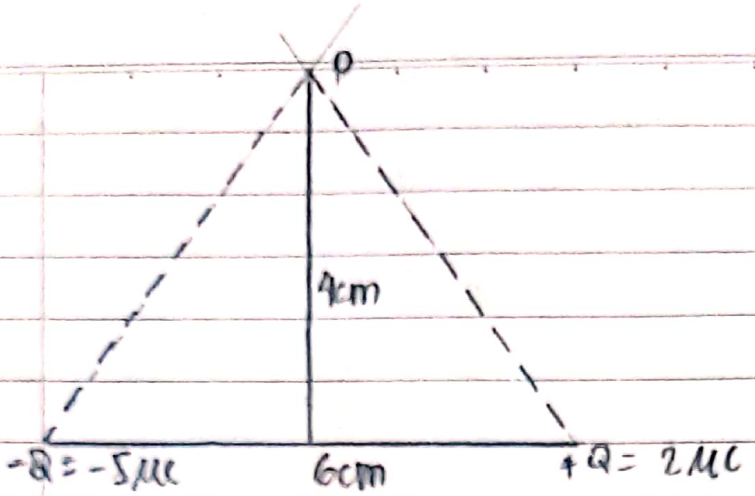
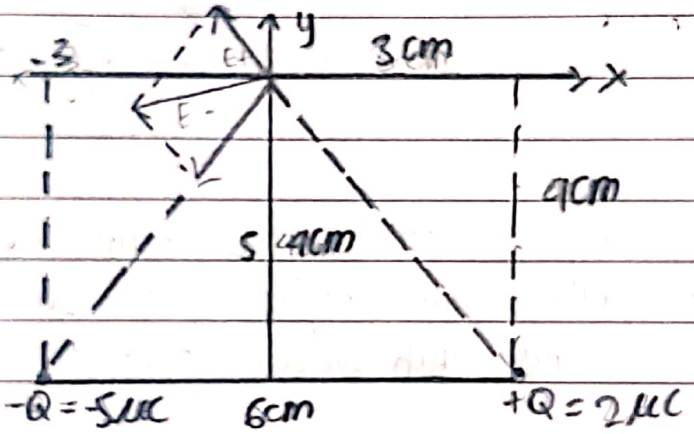
NATHAN SAMUEL PARLINDUNGAN
(2313499084)
1C - TNK

No.
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1.



terdapat dua muatan titik yang besarnya sama tetapi berlawanan tanda terpisah dengan jarak d, tentukan medan listrik pada titik P



$\bar{r}_+ = (3i - 4j)$ $\quad r = \sqrt{3^2 + 4^2}$

$= -3\hat{i} + 4\hat{j}$ $\quad = \sqrt{9 + 16}$

$= \sqrt{25} \Rightarrow 5$

$$E_1 = k_e \frac{q_+}{r^3} \bar{r}$$

$$= \frac{9 \times 10^9 \ (2 \times 10^{-6}) \ (-3i + 4j)}{(5 \times 10^{-2})^3}$$

$$= \frac{18 \times 10^3}{125 \times 10^{-6}} \ (-3i + 4j)$$

$$= \frac{18}{125} \times 10^9 (-3i + 4j) \ N/C$$

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No.

Date:

$$\vec{E}_- = k_e \frac{Q_-}{r^2}(-3i + 4j)$$

$$= \frac{9 \times 10^9 \,(.5 \cdot 10^{-6})}{(5 \times 10^{-2})^3}(-3i + 4j)$$

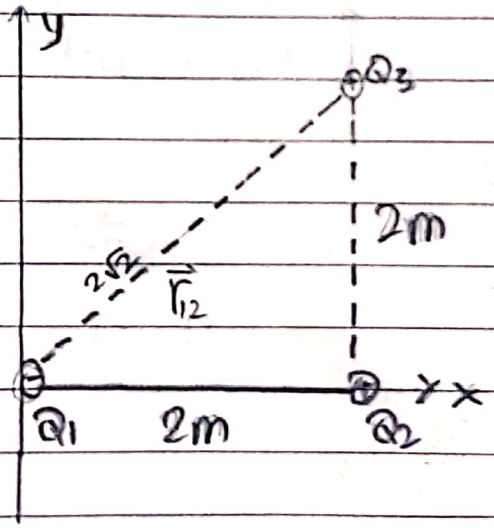
$$= \frac{45 \times 10^9}{125}(-3i + 4j)\ N/C$$

$$\vec{E}_p = E_+ + E_-$$

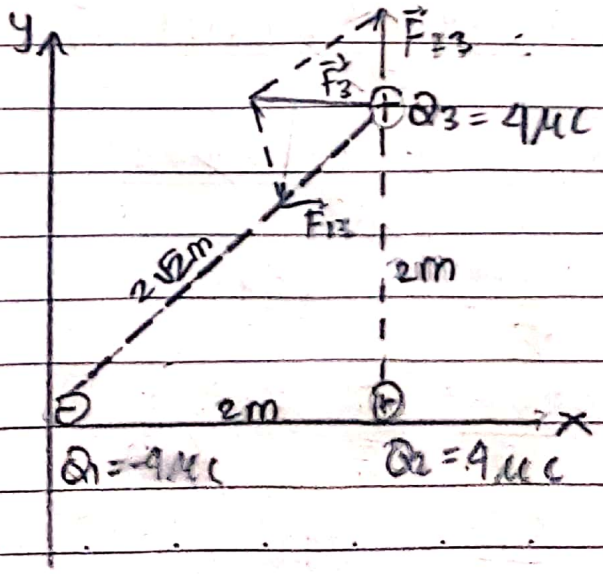
$$= \left(\frac{18}{125} + \frac{45}{125}\right)(-3\hat{i}) + \left(\frac{18}{125} + \frac{45}{125}\right)(4\hat{j}) \times 10^9\ N$$

$$= \left(\frac{63}{125}\right)(-3\hat{i}) + \left(\frac{63}{125}\right)(4\hat{j}) \times 10^9\ N$$

2.



Jawab:



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$r = \sqrt{2^2+2^2}$ $\quad \vec{r}_{23} = 2\hat{j}$

$r = \sqrt{4+4}$

$r = \sqrt{8}$

$$\vec{F}_{23} = k_e \frac{Q_1 Q_2}{(r_{23})^2} (2\hat{j})$$

$$\vec{F}_{23} = \frac{9\times10^9 \times 4\times10^{-6} \times 4\times10^{-6}}{2^3} (2\hat{j})$$

$$\vec{F}_{23} = \frac{9\times10^9 \times 10^{-6} \times 4\times10^{-6}}{2} (2\hat{j})$$

$$= \frac{36\times10^{-3}}{2}(2\hat{i}) = 18\times10^{-3}(2\hat{j})$$

$$= 36\times10^{-3}\hat{j}\,N //$$

$r = \sqrt{2^2+2^2}$

$r = \sqrt{4+4}$

$r = \sqrt{8}$

$\vec{r}_{13} = -(\vec{2}\hat{i} + \vec{2}\hat{j})$

$= -2\hat{i} - 2\hat{j}$

$$F_{13} = k_e \frac{Q_1 Q_3}{(r_{13})^3} \vec{r}_{13}$$

$$F_{13} = \frac{9\times10^9 \times 4\times10^{-6} \times 4\times10^{-6}}{(\sqrt{8})^3} (-\vec{2}\hat{i} - \vec{2}\hat{j})$$

$$F_{13} = \frac{9\times10^9 \times 16\times10^{-12} \times 16\times10^{-6}}{16\sqrt{2}} (-\vec{2}\hat{i} - \vec{2}\hat{j})$$

$$f_{13} = \frac{9\times10^{-3}}{\sqrt{2}} (-\vec{2}\hat{i} - \vec{2}\hat{j})$$

$$F_{13} = \frac{9\times10^{-3}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} (-2\hat{i} - 2\hat{j})$$

$$= \frac{9\sqrt{2}\times10^{-3}}{2}(-\hat{i} - \hat{j}) \Rightarrow 9\sqrt{2}\times10^{-3}[-\hat{i} - \hat{j})\,N //$$

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No.

Date:

$$F_3 = F_{13} + F_{23}$$

$$= 9\sqrt{2} \times 10^{-3} (-\hat{i} - \hat{j}) + 36 \times 10^{-3} \hat{j}$$

$$= -9\sqrt{2} \times 10^{-3} \hat{i} + (36 - 9\sqrt{2}) \times 10^{-3} \hat{j} \text{ N} //$$

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